

1. Literature Review

1.1 SAE Interpretability

In recent years, Sparse Autoencoders (SAEs) have emerged as a key tool for enhancing the interpretability of large models (Cunningham et al., 2023; Klindt et al., 2025). Their core idea is to decompose highly superimposed, polysemantic features in *fixed Transformer representations* into a set of *monosemantic basis vectors* via sparse reconstruction, thereby creating a *human-interpretable "feature-concept" dictionary*.

SAE demonstrates direct practical value in *steering scenarios*. Existing work shows that steering vectors constructed from SAE features can effectively modulate model behavior—without requiring additional supervisory data—for tasks such as safety alignment, role control, and reasoning path intervention (Arad et al., 2025; Chalnev et al., 2024; Wang et al., 2025). Compared to traditional fine-tuning or RLHF methods, this approach offers greater controllability and interpretability. Meanwhile, SAE concepts have been extended to multimodal and vision-language models to decompose representational structures in vision-content joint spaces (Pach et al., 2025). These advances collectively indicate that improving SAE decomposition quality is not only an interpretability challenge but also directly determines the *upper bound of model controllability*.

To achieve this goal, existing research predominantly focus on *refining SAE architectures* themselves, such as Gated SAEs (Rajamanoharan et al., 2024), Top-K SAEs (Gao et al., 2024), and Routed SAEs (Shi et al., 2025). These approaches enhance feature selection through gating, sparse selection, or dynamic routing mechanisms. Fundamentally, this direction improves SAEs' fitting capacity to pre-existing representations. However, a relatively underexplored avenue involves starting from the *Transformer's representation generation mechanism itself* to make it suitable for *posterior sparse decomposition*.

What-where decoupling is one approach to make sparse decompositions more suitable for posterior analysis (O'Neill et al., 2024). Chughtai et al. (2024) found that in the early layers of GPT-2 small (specifically, the *residual stream at layer 0*), positional and content features are almost completely decoupled. In this stage, sparse autoencoders (SAEs) can clearly recover the one-dimensional structure corresponding to positional encodings in the latent space, such as helical or periodic patterns. However, *as depth increases, what-where decoupling rapidly deteriorates*: positional information and semantic content are progressively superimposed within the residual stream, forming highly entangled representations, alongside more pronounced superposition and polysyntheticity (Cunningham et al., 2023; Klindt et al., 2025). In these deeper layers, SAEs typically require substantially larger dictionary sizes, stronger sparsity constraints, and even more

sophisticated post-alignment strategies to recover interpretable features (Karvonen et al., 2025; Minegishi et al., 2025).

However, it is crucial to emphasize that the entanglement between positional and content representations serves important purposes. Existing research shows that *what-where entanglement is a key source of the model's expressive power*. Bricken et al. (2023) argue that in finite-dimensional representation spaces, models inevitably rely on superposition—where different factors share representational subspaces—to efficiently encode numerous semantic features. Similarly, Su et al. (2021) demonstrate that appropriate positional-content coupling is essential for complex reasoning tasks. Thus, simply pursuing complete decoupling may compromise model performance.

Notably, *different positional encoding methods will lead to different what-where entanglement pattern* that significantly alter the spectral structure of attention logits (e.g., singular value distributions and approximate *Toeplitz properties*), thereby influencing information mixing patterns (Gu et al., 2025). For instance, multiplicative positional encodings like RoPE induce a degree of "spectral shrinkage," resulting in smoother attention patterns that *enhance training stability and generalization performance*. However, this smoothing can also enhance what-where entanglement, making posterior decomposition more challenging. In other words, by modifying the RoPE design, we can modulate what-where entanglement to optimize attention patterns—preserving RoPE's advantages while maximizing interpretability for posterior decomposition.

The need to balance detanglement and entanglement also becomes important in *multimodal scenarios*. In computer vision, *spatial position is a prerequisite for content*—pixel topology errors disrupt geometric relationships, so visual encoding must preserve tight entanglement between spatial structure and content representations (Yao et al., 2025; Chen et al., 2025). In contrast, positional information in LLMs exhibits greater independence: *the semantics of the same token vary little across positions* (Devlin et al., 2019). This endows language features with inherent decoupling potential. Multimodal models, however, face dual challenges: *fine-grained alignment between local visual regions and textual descriptions, and architectural coordination between textual priors and visual positional encodings* (Li et al., 2025; Huang et al., 2025).

This naturally leads to an appealing idea: if positional-content entanglement has an invertible structure, we could construct an exact bijection to transform entangled representations into decoupled ones without losing expressive power. This approach is formally analogous to normalizing flows, which achieve lossless separation of latent variables through invertible transformations (Kobyzev et al., 2019; Kingma & Dhariwal, 2018).

However, this idea is difficult to realize within the Transformer framework. First, multi-layer residual structures fundamentally perform information compression and redistribution rather than strict information preservation (Elhage et al., 2022). Second, superposition represents irreversible feature overlap in low-dimensional spaces, where distinct semantic factors exhibit lossy overlapping in their representations (Olah et al., 2020; Bricken et al., 2023).

Thus, positional-content entanglement in Transformers is inherently an irreversible mixture encoding. From this perspective, the true role of SAEs is to perform approximate sparse decomposition of mixed features within the given representation space, rather than strict factor decoupling (Cunningham et al., 2023; Gao et al., 2024). This suggests that, instead of pursuing theoretically infeasible complete reversible decoupling, a more practical direction is to develop sparsity-optimized decomposition strategies.

In other words, our goal is not to eliminate entanglement, but to reshape it: enabling representations that maintain model performance while achieving both efficient compression and posterior separability. By optimizing Transformer positional encodings, we can engineer positional-content coupling with more regular statistical structure, thereby reducing the difficulty of SAE-based approximate decoupling.

1.2 Positional Encoding

Current Transformer positional encoding schemes are rich and diverse, including Sinusoidal PE (Vaswani et al., 2017), Learned PE (Devlin et al., 2018; Dosovitskiy et al., 2020), T5-style relative biases (Raffel et al., 2020), ALiBi (Press et al., 2021), and RoPE (Su et al., 2021). RoPE, due to its excellent length extrapolation capability and training stability, has become the mainstream choice for SAE analysis (Cunningham et al., 2023; Gu et al., 2025; Zhang et al., 2026).

Mechanistically, RoPE achieves multiplicative positional-content coupling through complex-plane rotations, producing approximately Toeplitz attention patterns and exhibiting spectral contraction effects (Wang et al., 2025). This results in smoother attention distributions, enhancing training stability and positional generalization. Empirical studies also reveal clear functional differentiation among attention heads in RoPE models: some heads specialize in content matching, while others are more sensitive to relative positions (Gu et al., 2025).

However, this "functional differentiation" does not equate to true representational decoupling. RoPE implicitly embeds positional information through phase rotations, leaving position and semantics highly entangled in the representation space (Wang et al., 2025). Analyses show that RoPE's improvements primarily manifest in smoother mixing patterns, rather than fundamentally separating "what (content)" from "where (position)" (Gopalakrishnan et al., 2025). Thus, PoPE (Gopalakrishnan et al., 2025) proposes a more radical improvement: in polar coordinates, it explicitly decouples magnitude (content) from phase (position), structurally eliminating the what-where confound. Unlike RoPE's implicit coupling, PoPE maps the two to orthogonal geometric degrees of freedom, achieving clear division at the representation level.

Empirical validation shows that PoPE's structural adjustments not only preserve most of performance but deliver systematic improvements across position-sensitive tasks (perplexity, downstream metrics, zero-shot length extrapolation), consistently across model scales (124M–774M) (Gopalakrishnan et al., 2025). This indirectly confirms that redesigning positional-content coupling is both feasible and beneficial. However, the interaction mechanism between PoPE and SAEs remains unclear. Existing SAE research (Cunningham

et al., 2023; Klindt et al., 2025) treats positional encodings as fixed background, overlooking their potential as key variables for interpretability. A critical question remains unanswered: how does SAE decomposition behavior respond when the entanglement structure of underlying representations is altered through positional encoding changes?

From prior analysis, SAE decomposition of lossy mixture representations critically depends on feature separability. Thus, PoPE's structural decoupling is expected to significantly reduce SAE training difficulty, yielding:

- Improved feature monosemanticity
- Reduced reliance on large dictionaries and strong regularization
- Enhanced steering effectiveness

Meanwhile, SAE benchmarks (Karvonen et al., 2025) provide quantitative evaluation metrics. This paper therefore conducts the first systematic study of SAE performance under PoPE vs. RoPE, verifying whether positional encoding design—through decoupling positional and content features—can enhance interpretability while maintaining task performance.

I 2. Theoretical Investigations

Unlike computer vision, natural language exhibits the property of *translation invariance*: the core semantic meaning of a word generally remains consistent regardless of its position within a sentence. This suggests that Large Language Models (LLMs) should prioritize capturing relative positional dependencies over fixed absolute positions.

Both ALiBi and RoPE aim to eliminate the dependency on absolute positions within the attention mechanism, ensuring that attention scores are primarily driven by the relative relationships between tokens. While ALiBi achieves this via translational bias (applying linear penalties based on distance to the attention scores), RoPE employs rotational transformation (embedding relative position information directly into the vector space through geometric rotation). Unlike the original Transformer's sinusoidal positional encoding, which lacks an explicit mechanism for modeling relative distances, these approaches enable attention scores to be a function of the distance between tokens. This reliance on relative positioning is essential for improving long-range dependency modeling and is a key factor enabling the impressive length extrapolation capabilities observed in modern LLMs.

The primary limitation of RoPE lies in its implementation within the Cartesian coordinate system, where the 'what' (semantic content) and the 'where' (positional information) become inherently entangled through the rotation matrix operation. When calculating Query-Key matches, the model often struggles to disentangle whether the attention score is derived from semantic similarity or positional alignment. The derivation of RoPE is as follows:

1. We get word embedding input x with d dimensions in position p .

$$x^p = [x_1^p, x_2^p, \dots, x_d^p]$$

2. Then, split it into 2D vector.

$$x^p = (x_1^p, x_2^p, x_3^p, \dots, x_d^p) \Rightarrow (x_{1:2}^p, x_{3:4}^p, \dots, x_{d-1:d}^p)$$

3. We apply Rotation matrix R_θ for every group. The i_{th} group is as follow:

$$\begin{pmatrix} q_{2i-1}^p \\ q_{2i}^p \end{pmatrix} = R_\theta \begin{pmatrix} x_{2i-1}^p \\ x_{2i}^p \end{pmatrix} = \begin{pmatrix} \cos(p\theta_i) & -\sin(p\theta_i) \\ \sin(p\theta_i) & \cos(p\theta_i) \end{pmatrix} \begin{pmatrix} x_{2i-1}^p \\ x_{2i}^p \end{pmatrix}$$

$$\theta_i = 10000^{-2(i-1)/d}$$

4. We make it complex form. $\hat{i}$ means imaginary number. The i_{th} pair of dimension is:

$$x_i^p = (x_{2i-1}^p, x_{2i}^p) \Rightarrow (\text{complex form}) C_i^p = x_{2i-1}^p + \hat{i}x_{2i}^p$$

5. According to Euler's formula: $e^{i\theta} = \cos\theta + \hat{i}\sin\theta$, we can represent the the after rotation i_{th} pair as a complex multiplication:

$$C_i^{p'} = C_i^p \cdot e^{ip\theta_i}$$

6. Matrix R_θ could be regarded as rotating degree θ in complex space. The rotation parameter in complex space is:

$$e^{i\theta} = \cos\theta + \hat{i}\sin\theta$$

7. Expanding formula in part 3, we can get:

$$q_{2i-1}^p + \hat{i}q_{2i}^p = (x_{2i-1}^p + \hat{i}x_{2i}^p) \cdot (\cos(p\theta_i) + \hat{i}\sin(p\theta_i))$$

8. If complex $a + \hat{i}b = c + \hat{i}d$, then $a = c, b = d$, so can we obtain the rotated components:

$$q_{2i-1}^p = x_{2i-1}^p \cos(p\theta_i) - x_{2i}^p \sin(p\theta_i)$$

$$q_{2i}^p = x_{2i}^p \sin(p\theta_i) + x_{2i-1}^p \cos(p\theta_i)$$

where, x_{2i}^p represent what, $p\theta_i$ represent where, indicating "what - where" entanglement. this will lead to:

- Semantic Contamination: When performing attention calculations, the model must essentially 'undo' the rotation to extract the underlying semantic information and recover the original token embedding x . While theoretically trivial, this reversal process becomes computationally non-trivial as dimensionality scales; the model incurs additional overhead to disentangle the intertwined semantic and positional features.
- Informational Redundancy: Positional information persists as a persistent 'perturbation' across every dimension of the vector. During the computation of Query-Key dot products, this introduces significant 'noise'—irrelevant to semantic matching—that is purely positional in nature, thereby cluttering the attention scores with extraneous signals."

PoPE represents each scalar channel as a complex number in *polar form*, where the *magnitude* carries content and the *phase* carries position.

1. We get word embedding input x with d dimensions at position p :

$$x^p = [x_1^p, x_2^p, \dots, x_d^p]$$

2. Unlike RoPE, PoPE does not group every two dimensions into a 2D Cartesian vector. Instead, it treats each scalar feature as one channel and converts it into the *magnitude* of a complex number:

$$x^p = [x_1^p, x_2^p, \dots, x_d^p]$$

3. PoPE maps each scalar channel into a non-negative magnitude using the softplus function $\sigma(x) = \ln(1 + e^x)$, which allows that value to be interpreted as a magnitude in

polar coordinates. μ means magnitude.

$$\mu_i^p = \sigma(x_i^p) = \ln(1 + e^{x_i^p})$$

4. PoPE assigns position only to the phase. For the i_{th} channel, the phase is determined purely by the token position p and the channel frequency θ_i , so content no longer shifts the phase. ϕ means phase

$$\phi_i^p = p\theta_i$$

$$\theta_i = \theta^{(i-1)/d} \text{ (} \theta \text{ usually use 10000)}$$

5. Now each channel is written directly in polar-complex form. Its magnitude μ_i^p stores content, and its phase ϕ_i^p stores position

$$q_i^p = \mu_i^p e^{i\phi_i^p} = \mu_i^p e^{ip\theta_i}$$

6. According to Euler's formula, the complex number can be expanded into Cartesian coordinates, but unlike RoPE, this expansion starts from a magnitude-phase representation rather than from a 2D semantic pair

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$q_i^p = \mu_i^p (\cos(p\theta_i) + i \sin(p\theta_i))$$

If we write the real and imaginary parts explicitly, they are obtained from the same magnitude μ_i^p and the same position phase $p\theta_i$. This is the key structural difference from RoPE: the semantic quantity enters only through the magnitude, not through the phase offset.

Although PoPE elegantly disentangles the 'what' and 'where' components, it introduces several limitations compared to RoPE:

- **Lack of Hardware/Kernel Compatibility:** RoPE is deeply integrated into the current software ecosystem. Mainstream attention kernels (e.g., FlashAttention) are highly optimized for the RoPE-based rotation operator. Transitioning to PoPE would necessitate the development of custom, hardware-optimized kernels, resulting in significant deployment and integration costs.
- **Computational Overhead:** The requirement for per-channel non-linear activation (e.g., Softplus) introduces extra arithmetic operations for every token. While seemingly marginal, this increases the computational burden on the GPU's arithmetic logic units during the attention forward/backward pass.
- **Over-constrained Inductive Bias:** PoPE's strict decoupling of content and position may be counter-productive in complex reasoning tasks. By imposing a rigid geometric separation, the model may be deprived of subtle what-where interactions that are often essential for high-level semantic reasoning, potentially creating an 'upper bound' on performance in tasks requiring intricate logic. *And that's what I want to make a difference.*

| 3. Problem Definition

RQ1: Performance Stability

Does PoPE outperform RoPE in language model training and downstream tasks?

RQ2: Decoupling Effectiveness

Does PoPE achieve stronger content-position decoupling?

RQ3: Interpretability Improvement

Does PoPE enable SAEs to learn more monosemantic features?

RQ4: Mechanistic Impact

How does positional encoding affect attention logits' spectral structure (Toeplitz properties)?

Then to conclude:

Key Hypotheses to Test

1. Does what-where disentanglement harm language model training efficiency? If yes, how?
2. Does what-where disentanglement yield purer semantic features in SAE training? If yes, how?

This systematic comparison will clarify PoPE's practical value and inspire future positional encoding designs.

I 4. Task Flow in progress

Phase 1: Model Construction (Weeks 1-3) *Completed*

Limited by computing resources, we only train a clean 90M Toy Transformer from scratch with switchable RoPE/PoPE implementations to validate pipeline feasibility and test PoPE's positional processing hypothesis.

Remark: I planned to monkey patch Llama model and inject PoPE into it. But different PE will lead to very bad performance.

```
# Data Preparation
DATA_CFG = SimpleNamespace(
    dataset = "wikitext",
    config = "wikitext-103-raw-v1",
    key = "text",
    raw_text_samples = 30000,
    train_blocks = 1500,
    valid_blocks = 300,
    seq_len = 1024,
    tokenizer = "gpt2",
    eos_join = False,
    seed = 42,
)

DATA_RES = {
    vocab_size = 50257,
    total_token_used = 1891328,
    raw_text = 30000,
    non_empty_texts = 19357,
```

```

used_texts = 14926,
}

```

```

# Model Construction
MODEL_CFG = SimpleNamespace(
    n_layers = 12,
    d_embed = 512,
    n_heads = 8,
    d_ff = 2048,
    dropout = 0.1,
    seq_len = DATA_CFG.seq_len,
    # Extrapolation test is not being considered at this time.
    lr = 3e-5,
    vocab_size = DATA_RES["vocab_size"]
)

MODEL_RES = {
    RoPE_para = 89343057, # about 90M
    PoPE_para = 89346129, # about 90M
}

```

Phase 2: Train & Evaluate (week3-6) *Completed*

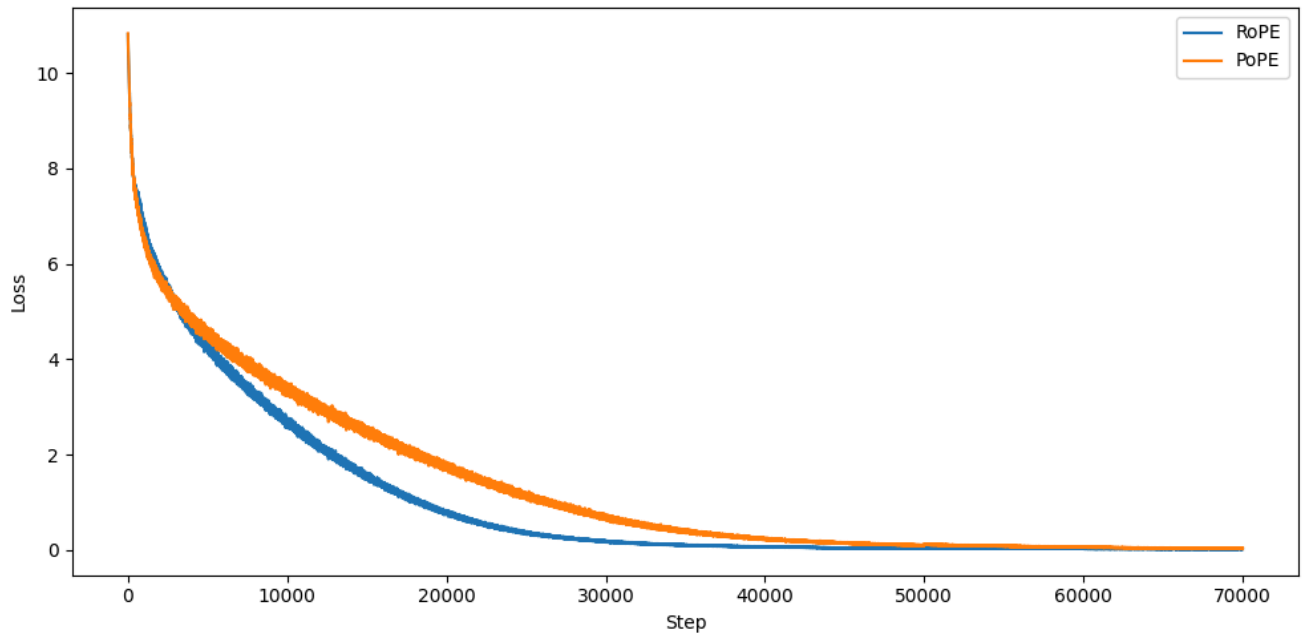
- Stability Analysis: Compare training loss curves and gradient stability between the two architectures.
- Performance Evaluation: Benchmark models on language modeling (perplexity) and zero-shot length extrapolation tasks.

```

TRAIN = SimpleNamespace (
    steps = 70000, #20000比较好
    batch_size = 24,
    seq_len = MODEL.seq_len,
    save_interval = 10000,
    log_interval = 200,
    data_type = "bfloat16",
    warmup = 2000
)

```


Loss Curve:



model	layer	attn_entropy	avg_attn_distance
rope	0	5.83	243.78
pope	0	5.79	187.10
rope	1	2.83	238.01
pope	1	1.78	27.77
rope	2	4.12	205.83
pope	2	3.18	31.61
rope	3	4.91	189.53
pope	3	3.56	41.66
rope	4	4.97	172.63
pope	4	3.55	49.17
rope	5	4.75	147.42
pope	5	3.95	58.54
rope	6	4.67	132.54
pope	6	3.90	59.48
rope	7	4.35	116.70
pope	7	4.35	67.65
rope	8	4.69	119.71
pope	8	4.44	66.49
rope	9	4.63	109.58
pope	9	4.55	74.17
rope	10	4.50	95.96
pope	10	4.58	75.78
rope	11	4.63	108.91
pope	11	4.32	68.81

1. **attn_entropy**: Compute the Shannon entropy of each query's attention distribution by treating its normalized attention weights over all keys as a probability distribution. A lower entropy indicates that attention is more concentrated on a few positions, while a higher entropy indicates a more dispersed attention pattern.

$$H = - \sum_{i=1}^N a_i \log a_i$$

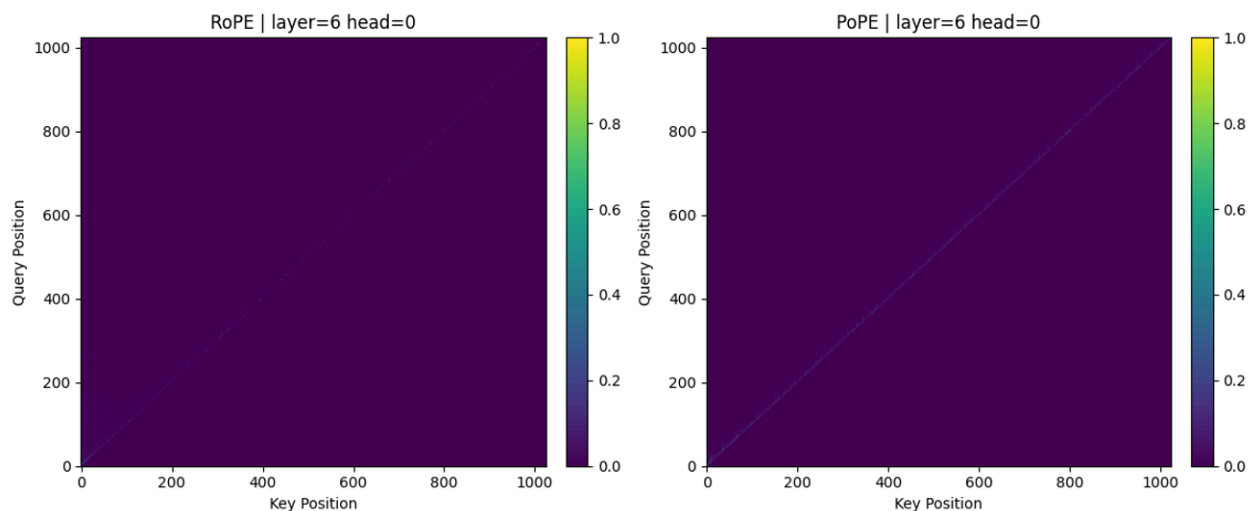
a_i is the attention weight at position i , N is the number of tokens paid attention to.

2. **avg_attn_distance**: Compute the attention-weighted average token distance, which reflects the model's average contextual span. A larger value indicates a stronger preference for long-range dependencies, whereas a smaller value indicates a stronger preference for local dependencies.

$$Distance = \sum_{i=1}^N a_i d_i$$

d_i is the distance between query word and the key word

3. **attn_heatmap**: High responses near the diagonal usually indicate that the model tends to focus more on adjacent or nearby tokens.



Phase 3: SAE Training & Core Metrics (Weeks 6-8) *Partly Completed*

Train SAEs (not top-k SAE) with identical configurations on matched RoPE/PoPE models across early/mid/late layers. Collect metrics as follow:

- Reconstruction Error
- L0 Norm
- Dead Features

```
SAE_CFG = SimpleNamespace(  
    activation layer = "residual after block"  
    dictionary size = 4096  
    L1 coefficient = 1e-2  
    epochs = 400  
    learning rate = 2e-3  
    batch size = 2048  
    validation row cap = 8192 # sampling size of SAE validation set affects  
    stability of MSE/nz.  
)
```

Model	Val MSE	L0-Norm rate
RoPE	0.0261	8.59%
PoPE	0.0216	6.59%

From the current simple metrics, PoPE shows relatively low ValMSE and L0-Norm, which to some extent supports our hypothesis.

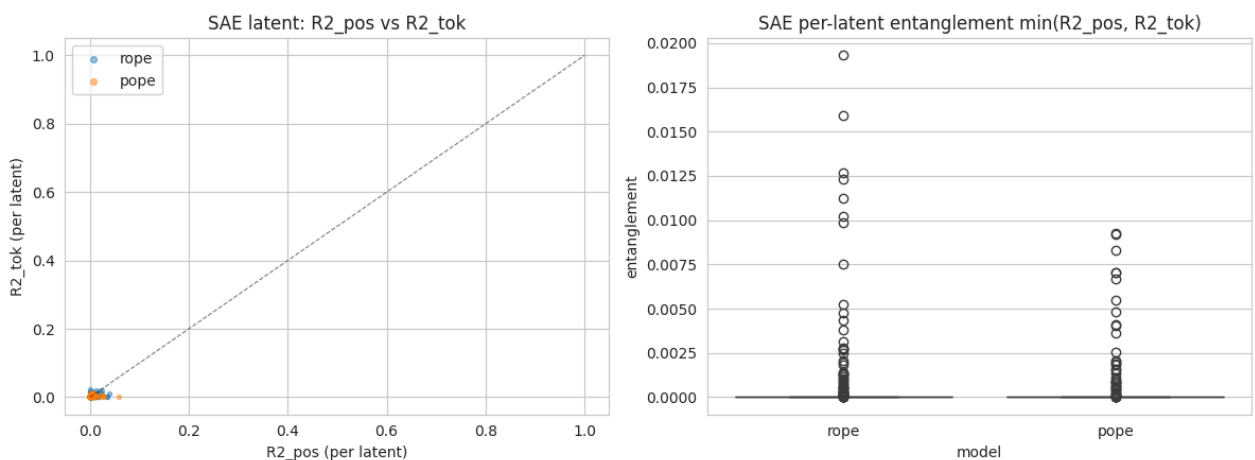
model	layer	mean_abs_corr	max_abs_corr	mean_mi	max_mi
rope	2	0.010157	0.997380	0.000437	0.136369
rope	6	0.006317	0.948309	0.000297	0.066435
rope	10	0.009525	0.998062	0.000383	0.053764
pope	2	0.013409	0.993653	0.000555	0.090393
pope	6	0.008905	0.937369	0.000400	0.039391
pope	10	0.007532	0.814278	0.000348	0.032652

- **Correlation:** measures the degree of linear relationship between features; lower values indicate weaker linear entanglement.
- **Mutual Information:** measures the mutual information between features; lower values indicate weaker nonlinear dependence and a greater degree of independence.

Phase 4: Disentanglement Benchmark (Weeks 9-11) *Partly Completed*

According to [Bilal Chughtai et al. \(2024\)](#), we use the Disentanglement Probe test, there are several important metrics as follow:

- R^2_{tok} : Indicates how single SAE feature could predict the semantic
- R^2_{pos} : Indicates how single SAE feature could predict the position



Phase 5: LLM-Assisted SAE Latent Interpretation (Weeks 11-13) *Partly Completed*

After each TopK-SAE training run, we use GPT-4o to automatically interpret all SAE latents and use the outputs as a quantitative measure of interpretability. Specifically, for each latent, we provide approximately 100–200 activation contexts together with their activation strengths and token positions, then use a fixed prompt to have GPT-4o generate a brief

natural-language explanation of its activation pattern. Each latent is evaluated three times and the results are averaged to reduce randomness. The GPT-4o outputs are then used to assess the interpretation quality of each latent along the following three dimensions:

1. *Monosemanticity*: whether the latent corresponds to a single, clear concept.
2. *Low-level vs. high-level feature type*: whether the latent is more oriented toward low-level syntax, punctuation, or part-of-speech features versus high-level semantics, topics, or intents.
3. *Explanation examples*: whether the examples are representative enough to cover activation patterns across multiple different contexts.

Mainly refer work of [Paulo et al., \(2024\)](#) and [Shi et al., \(2025\)](#). Output samples are as follow:

```
gpt-4o interpret result:
```

```
AVG Monosemanticity Score: 3.95/5.00
AVG Low-level Features Score: 4.14/5.00
AVG High-level Features Score: 4.57/5.00
API Cost: $0.4261
```

```
Examples of Interpreted Features:
```

```
Feature #5: low-level (Score:5) - The activations are consistently
associated with relative pronouns "which" and "that," indicating a ...
Feature #15: low-level (Score:4) - The activations are consistently
associated with punctuation marks, particularly those used in dialo...
Feature #23: low-level (Score:4) - The activations are consistently
associated with the token "than," which suggests a pattern related ...
```

I have also started using GPT-4o to help interpret features, but the code for a rigorous quantitative evaluation is not yet complete, so for now I will only present the following cases.

I 5. Future Task flow

Phase 6: Strengthening Experimental Reliability (Weeks 13-16)

1. Expand the experiments to larger datasets in contexts of varying complexity to test if what-where disentanglement will affect the learning ability in complex context?
2. Validate the results across multiple random seeds.
3. Introduce additional dynamic monitoring metrics to track representational changes, including:
 - SAE feature reuse rate
 - Monosemanticity growth
 - Dead feature evolution

Phase 7: Steering and Causal Intervention (Weeks 16-20)

The final phase evaluates the practical utility of the disentangled features. We will construct steering vectors from SAE features to intervene in model behavior, such as controlling

safety-related or stylistic outputs. We hypothesize that steering vectors derived from PoPE-based SAEs will achieve higher precision and fewer unintended side effects, due to the cleaner underlying feature space.

Phase 8: New Positional Embedding Design (Exploratory)

The insights from the preceding analyses may guide the design of a new positional embedding scheme that preserves strong representational capacity during learning while remaining more amenable to approximate decoupling in SAE analysis. Or combine different what-where extent, using different positional encoding for different head. (It's still just an imagination up to now; its feasibility hasn't been verified.)

I guess it may be helpful for multimodal tasks. But I need to read more and test more to verify my conjecture

I 6. Risk Mitigation

A potential risk is the trade-off between strict disentanglement and model expressivity. Should PoPE show performance degradation in complex reasoning tasks, we will investigate hybrid coupling strategies or partial disentanglement to strike an optimal balance between interpretability and model capacity.

I 7. Conclusion

This research project re-conceptualizes mechanistic interpretability not merely as a posterior analysis task, but as a holistic problem requiring the co-design of model architectures and interpretability tools. By demonstrating that PoPE significantly lowers the burden of SAE feature decomposition, this work paves the way for more robust and controllable Large Language Models and inspire creation of more powerful PE with higher SAE compatibility.

I 8. Appendix

1. [Cunningham et al., 2023, "Sparse Autoencoders Find Highly Interpretable Features in Language Models"](#)
2. [Klindt et al., 2025, "Towards Monosemanticity in Neural Networks"](#)
3. [Klindt et al., 2025, "From superposition to sparse codes: interpretable representations in large language models"](#)
4. [Arad et al., 2025, "SAEs Are Good for Steering – If You Select the Right Features"](#)
5. [Chalnev et al., 2024, "Improving Steering Vectors by Targeting Sparse Autoencoder Features"](#)
6. [Wang et al., 2025, "Improving LLM Reasoning through Interpretable Role-Playing Steering \(SRPS\)"](#)
7. [Pach et al., 2025, "Sparse Autoencoders Learn Monosemantic Features in Vision-Language Models"](#)
8. [Rajamanoharan et al., 2024, "Improving Dictionary Learning with Gated Sparse Autoencoders"](#)

9. [Gao et al., 2024, "Scaling and evaluating sparse autoencoders"](#)
10. [Shi et al., 2025, "Route Sparse Autoencoder to Interpret Large Language Models"](#)
11. [O'Neill et al., 2024, "Disentangling Dense Embeddings with Sparse Autoencoders"](#)
12. [Chughtai et al., 2024, "Understanding positional features in layer 0 SAEs"](#)
13. [Karvonen et al., 2025, "SAEBench: A Comprehensive Benchmark for Sparse Autoencoders in Language Model Interpretability"](#)
14. [Minegishi et al., 2025, "Rethinking Evaluation of Sparse Autoencoders through the Poly-Semantic Evaluation Framework"](#)
15. [Gu et al., 2025, "Unpacking Positional Encoding in Transformers: A Spectral Analysis of Content-Position Coupling"](#)
16. [Bricken et al., 2023, "Towards Monosemanticity: Decomposing Language Models via Superposition"](#)
17. [Su et al., 2021, "RoFormer: Enhanced Transformer with Rotary Position Embedding"](#)
18. [Yao et al., 2025, "GeoPE: A Unified Geometric Positional Embedding for Structured Tensors"](#)
19. [Chen et al., 2025, "A 2D Semantic-Aware Position Encoding for Vision Transformers"](#)
20. [Devlin et al., 2018, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding"](#)
21. [Vaswani et al., 2017, "Attention Is All You Need"](#)
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